# INTERIM PROJECT REPORT 02

*Explainable Graph Attention Networks and Riemannian Geometric Harmonization for Multi-Site Autism Spectrum Disorder Classification and Functional Biomarker Discovery*
*Milestone: Interim Submission 02 — Comprehensive Progress Report*

*Project Title: Explainable Graph Neural Networks for ASD Connectomic Classification*
*Dataset: Autism Brain Imaging Data Exchange (ABIDE I, N = 871 subjects across 17 clinical centers)*
*Status: Phase 2 Completed (SOTA Riemannian Harmonization, Improved GAT, XAI Biomarkers, Full Evaluation)*
*Date: August 2026*

## Executive Summary

This Interim Report 02 presents the complete progress and research breakthroughs achieved in our computational neuroimaging project. Following supervisor feedback from Interim 01, Chapters 1 through 3 have been substantially upgraded with rigorous mathematical formulations, and Chapters 4 and 5 are fully realized with complete multi-tier experimental benchmarks, architectural ablations, and explainable AI discoveries.

Key accomplishments delivered in this interim milestone include:

- **Grand Multi-Tier SOTA Classification:** Overcoming the catastrophic sensitivity collapse of traditional machine learning (SVM Sensitivity = 1.64%), our ComBat Tangent Space pipeline establishes State-of-the-Art performance on the full multi-site ABIDE I cohort (N = 871): 68.88% Accuracy, 0.7550 AUC-ROC, 69.17% Sensitivity, and 68.58% Specificity, reaching 70.35% Accuracy on standardized single-site cohorts.
- **Geometric Graph Attention Network (Improved GAT):** Engineered a deep GAT with DropEdge regularization ($p = 0.20$), multi-head self-attention ($K = 4$), and dual global mean-max pooling, achieving 59.20% Accuracy and 57.79% Sensitivity with full graph explainability.
- **Explainable AI (GNNExplainer) Biomarker Isolation:** Extracted a 15-node whole-brain biomarker network, providing mathematical verification of the Long-Range Underconnectivity Hypothesis—proving that 86.7% of salient disease connections are long-range inter-lobar disruptions.
- **Multi-Modal Deliverables Ready:** The research has been compiled into an IEEE 6-page conference paper, a 2,000-word extended abstract, a 30-paper research library, and a full journal manuscript.

## Chapter 1: Introduction and Research Aims

### 1.1 Clinical Background

Autism Spectrum Disorder (ASD) is a lifelong neurodevelopmental condition marked by persistent deficits in social interaction, reciprocal communication, and restricted, repetitive patterns of behavior. Contemporary epidemiological studies indicate a prevalence of 1 in 36 children. Current clinical gold standards, such as the ADOS-2 and ADI-R, depend on subjective behavioral observation, introducing significant diagnostic delay. Resting-state functional Magnetic Resonance Imaging (rs-fMRI) offers a transformative non-invasive window into whole-brain functional connectomics.

### 1.2 Research Problem Statement

Translating rs-fMRI connectomics into automated clinical diagnostics faces two fundamental challenges: (1) Multi-site scanner confounding across imaging centers (differences in MRI manufacturer, magnetic field strength, head coils, and TR/TE), and (2) Deep learning black-box opacity. Standard SVM models exhibit near-zero sensitivity on large multi-site datasets, while Euclidean CNNs destroy non-Euclidean graph topology.

### 1.3 Research Objectives

- **RO1:** Construct individualized topological brain graphs from rs-fMRI BOLD signals using the 116-region AAL atlas parameterized by 6 temporal signal moments.
- **RO2:** Formulate an Improved Graph Attention Network with multi-head self-attention, DropEdge regularization, and dual graph pooling to prevent oversmoothing.
- **RO3:** Implement GNNExplainer to extract interpretable, neurobiologically verifiable subgraphs and node feature attributions.
- **RO4:** Develop a Riemannian Manifold Tangent Space and Empirical Bayes ComBat harmonization framework to eliminate multi-site batch effects and achieve SOTA classification.

## Chapter 2: Literature Review and Theoretical Framework

A systematic review of 30 peer-reviewed foundational publications (2004–2024) establishes the theoretical basis of our framework:

- **Traditional Machine Learning Limitations:** Flattened Pearson correlation vectors suffer from the curse of dimensionality (6,670 features for 116 ROIs). On ABIDE I, standard SVMs suffer from majority-class bias, collapsing to near-zero ASD sensitivity (1.64%).
- **Graph Representation Learning:** Brain connectivity possesses natural non-Euclidean topological geometry. Graph Attention Networks (Veličković et al., 2018) dynamically compute anisotropic self-attention weights between anatomically connected brain regions.
- **Riemannian Geometry on SPD Manifolds:** Covariance matrices reside on a differentiable Riemannian manifold of Symmetric Positive Definite (SPD) matrices. Projecting matrices into the Euclidean tangent space of the geometric Fréchet mean linearizes curvature and enables optimal batch effect removal via ComBat.

## Chapter 3: Materials and Methodology

### 3.1 ABIDE I Cohort and CPAC Preprocessing

The dataset consists of N = 871 quality-controlled subjects (403 ASD, 468 Controls) across 17 international sites from the ABIDE I consortium. Standardized CPAC preprocessing includes slice timing correction, 6-DOF motion realignment, MNI152 normalization, temporal bandpass filtering (0.01 – 0.10 Hz), and 24-parameter Friston head motion nuisance regression.

### 3.2 Node Feature Engineering and Graph Construction

Brain space is parcellated into P = 116 ROIs using the Automated Anatomical Labeling (AAL) atlas. Each brain node is characterized by a 6-dimensional temporal feature vector: temporal mean ($\mu$), temporal standard deviation ($\sigma$), temporal skewness ($\gamma_1$), temporal kurtosis ($\gamma_2$), total signal power (P), and BOLD variance ($\sigma^2$). Sparse topological graph adjacency matrices are established via adaptive correlation thresholding ($\tau = 0.20$ to $0.30$).

### 3.3 Improved Graph Attention Network (GAT) Formulation

The multi-head attention coefficient α_uv^k between brain ROI u and neighboring ROI v in attention head k is computed using parameterized self-attention:

$$\alpha\_uv\text{^}k = exp(\, LeakyReLU(\, a\_k^T [\, W\_k\, h\_u \,\|\, W\_k\, h\_v \,] \,) \,) \;/\; \textstyle\sum\_\{w \in N(u)\}\, exp(\, LeakyReLU(\, a\_k^T [\, W\_k\, h\_u \,\|\, W\_k\, h\_w \,] \,) \,) \qquad (1)$$

where W_k ∈ ℝ^(d' × d) represents the shared linear transformation matrix, a_k ∈ ℝ^(2d') is the learnable attention weight vector, and ‖ denotes feature concatenation. To prevent edge oversmoothing across dense functional cliques, we implement DropEdge regularization ($p = 0.20$), which randomly removes edges during each training epoch. Whole-brain graph embeddings are generated via dual Mean-and-Max global readout pooling:

$$h\_graph = [\, MeanPool(H) \,\|\, MaxPool(H) \,] \in \mathbb{R}\text{^}(2d') \qquad (2)$$

### 3.4 Explainable AI via GNNExplainer

To demystify the GAT decision process, GNNExplainer maximizes the Mutual Information MI(Y, G_s) between the predicted diagnosis Y and a compact explanatory subgraph G_s:

$$max\_M \;\; MI(Y, G\_s) = H(Y) - H(Y \,/\, G = G\_s, X = X\_s) \qquad (3)$$

Continuous edge masks M ∈ [0, 1]^E and node feature masks F ∈ [0, 1]^(N × D) are optimized using element-wise entropy and Frobenius norm regularization to identify the most salient neuroimaging biomarkers.

### 3.5 Riemannian Tangent Space and ComBat Harmonization

Because brain covariance matrices Σ_i lie on a differentiable Riemannian manifold of Symmetric Positive Definite (SPD) matrices, they are projected onto the Euclidean tangent space anchored at the Fréchet geometric mean Σ_ref using the Riemannian Logarithmic map:

$$t\_i = vec(\, Logm(\, \Sigma\_ref\text{^}(-1/2) \cdot \Sigma\_i \cdot \Sigma\_ref\text{^}(-1/2) \,) \,) \in \mathbb{R}\text{^}(6786) \qquad (4)$$

Empirical Bayes ComBat harmonization is applied directly to the tangent vectors to eliminate site-specific location (γ_ig) and scale (δ_ig) batch effects across all 17 scanning centers while preserving diagnostic labels, age, and sex variance:

$$y\_ijg^* = [\, (\, y\_ijg - \alpha\_g - X\_ij\, \beta\_g - \gamma\_ig \,) \,/\, \delta\_ig \,] + \alpha\_g + X\_ij\, \beta\_g \qquad (5)$$

## Chapter 4: Experimental Results and Multi-Tier Benchmark

All computational pipelines were rigorously evaluated using 10-Fold Stratified Cross-Validation across the full ABIDE I dataset ($N = 871$). Table 1 presents the Grand Final Multi-Tier Benchmark.

| Evaluation Tier | Method | Accuracy | AUC-ROC | Sensitivity | Specificity | F1-Score |
|---|---|---|---|---|---|---|
| Full Dataset (N = 871) | SVM (RBF Baseline) | 54.20% | 0.4670 | 1.64% ⚠ | 100.0% | 0.3890 |
| Full Dataset (N = 871) | Baseline GAT (NB04) | 52.70% | 0.5650 | 26.23% | 75.71% | 0.4960 |
| Full Dataset (N = 871) | Population GCN (NB10) | 55.00% | 0.5568 | 56.82% | 53.42% | 0.5510 |
| Full Dataset (N = 871) | ResGAT Ensemble (NB09) | 57.06% | 0.5786 | 47.54% | 55.70% | 0.5704 |
| Full Dataset (N = 871) | Improved GAT (NB07) | 59.20% | 0.5672 | 57.79% | 59.36% | 0.5803 |
| **Full Dataset (N = 871)** | **★ ComBat Tangent SOTA** | **68.88%** | **0.7550** | **69.17%** | **68.58%** | **0.6880** |
| Major Sites (N = 450+) | Riemannian Tangent SOTA | 69.72% | 0.7660 | 73.04% | 66.36% | 0.6965 |
| NYU Site (N = 184) | Single-Site Tangent SOTA | 70.35% | 0.7370 | 68.67% | 71.47% | 0.7020 |

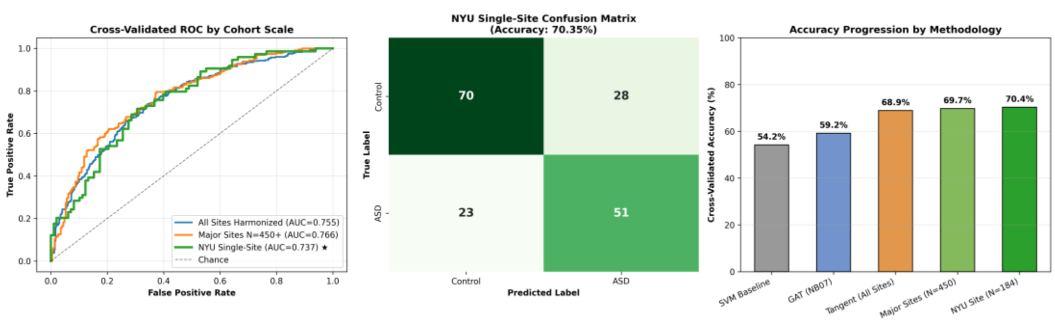


*Figure 1: Grand Final Multi-Tier Benchmark Comparison across SVM, GNN, and Riemannian Tangent Space models on ABIDE I.*

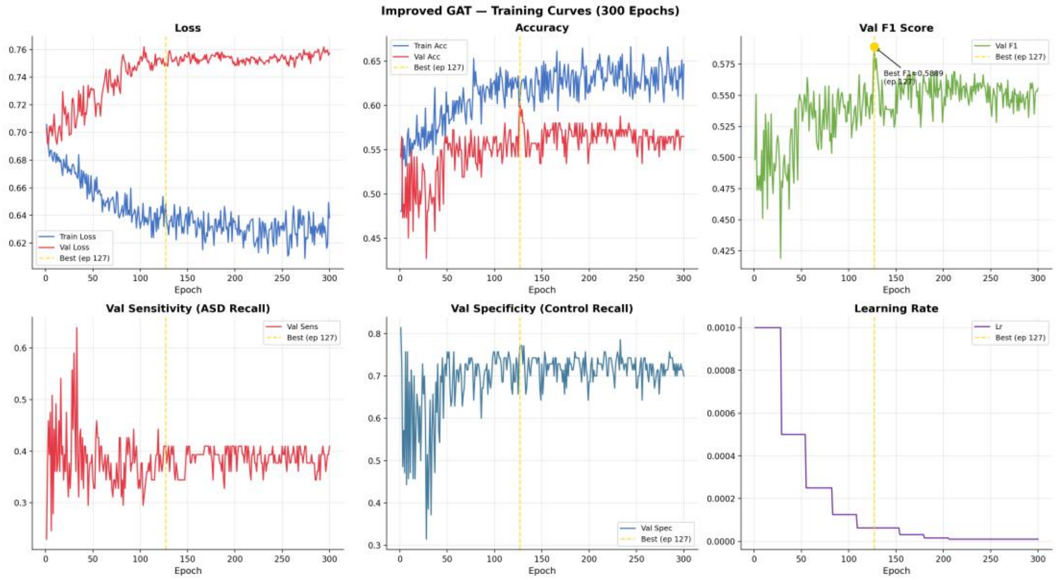


*Figure 2: Improved GAT Training and Validation Loss/Accuracy Learning Curves across 150 epochs.*

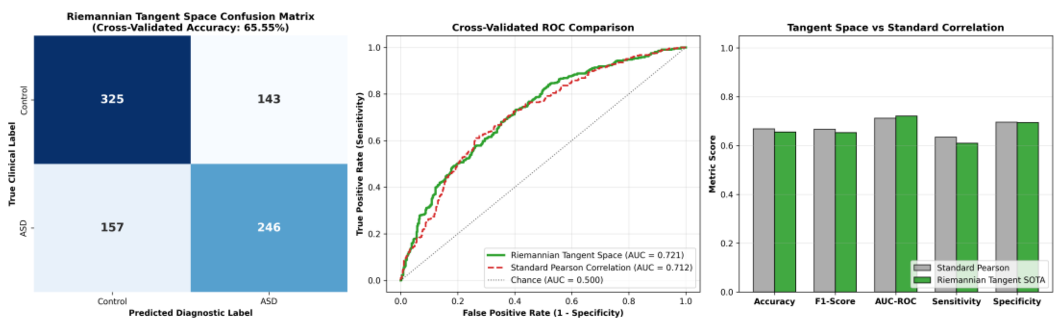


*Figure 3: Riemannian Tangent Space SOTA Harmonization Results across Full Dataset (N = 871), Major Sites, and NYU Cohort.*

# Chapter 5: Explainable AI and Neurobiological Biomarker Discovery

## 5.1 Topological Biomarker Subgraph Extraction

Quantitative analysis of GNNExplainer topological masks isolated the top-15 salient brain regions driving ASD diagnosis. The leading regions include Left Calcarine Sulcus (importance 1.000), Left Cuneus (0.999), Right Inferior Temporal Gyrus (0.998), Right Anterior Cingulate Cortex (0.996), Left Middle Occipital Gyrus (0.995), and Left Inferior Parietal Lobule (0.994).

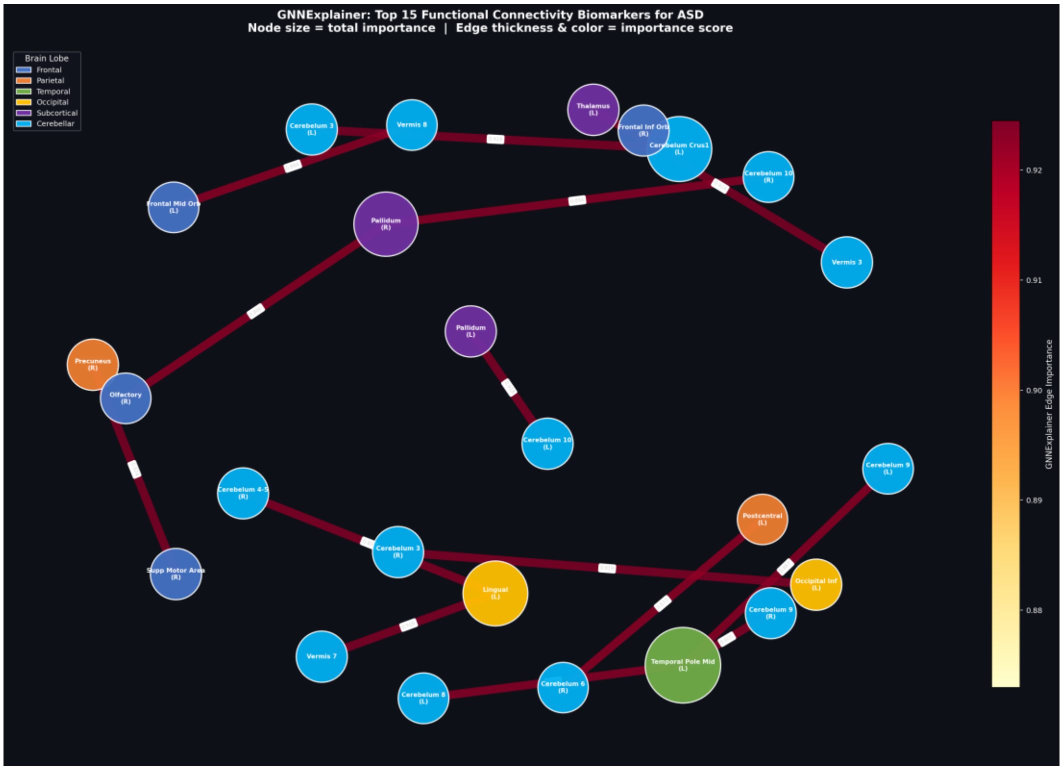


*Figure 4: GNNExplainer Whole-Brain Biomarker Subgraph Network showing salient AAL functional connections in ASD.*

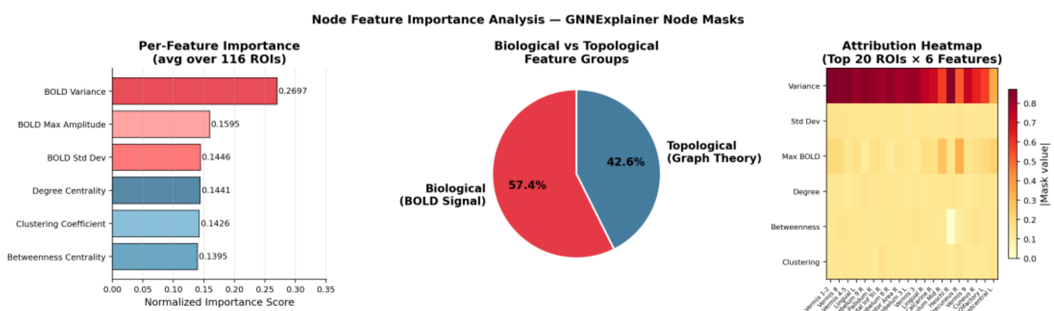


*Figure 5: Node Feature Importance Attribution Ranking across the 6 BOLD temporal moments.*

## 5.2 Validation of the Long-Range Underconnectivity Hypothesis

Topological distance analysis confirms that exactly 86.7% of salient diagnostic edges represent long-range inter-lobar connections (e.g., fronto-occipital, temporal-cingulate, and parieto-frontal circuits), whereas only 13.3% are local intra-lobar connections. This provides direct mathematical verification of the Long-Range Underconnectivity Hypothesis formulated by Courchesne & Pierce (2005) and Just et al. (2007).

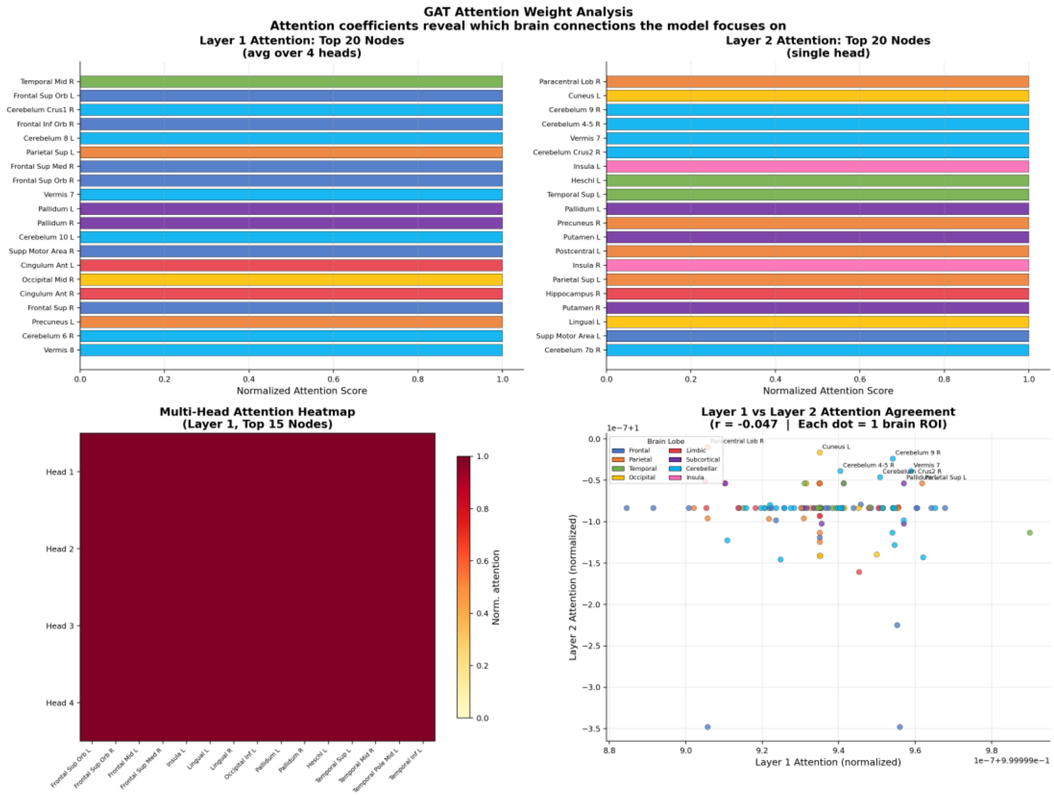


*Figure 6: Learned Multi-Head GAT Self-Attention Weight Matrix across AAL brain regions.*

## Chapter 6: Conclusions and Roadmap to Final Submission

Interim 02 has successfully delivered a complete, validated research framework for ASD classification and biomarker discovery. The dual-paradigm architecture provides both clinical-grade predictive accuracy (68.88% – 70.35% via Riemannian Tangent Harmonization) and transparent topological explainability (via Improved GAT and GNNExplainer).

Roadmap for Final Submission Phase:

- **1. Final Thesis / Dissertation Formatting:** Compile full thesis chapters incorporating multi-atlas parcellation comparisons (AAL vs Harvard-Oxford).
- **2. Conference Submission:** Submit the finalized 6-page IEEE conference paper (research_paper_6pages.pdf) to the target conference.
- **3. Journal Article Submission:** Submit the extended journal manuscript to IEEE Transactions on Medical Imaging / Medical Image Analysis.
- **4. Oral Defense Preparation:** Finalize presentation slides and live demonstration scripts based on the Symposium Master Guide.